

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of M. Sc. (AOC) Examination March 2018

OC 722: Concepts in Organic Chemistry

Date: 29/03/2018 Day: Thursday Time: 10.00 a.m. to 10.30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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| Q-I | Choose the correct answer for the following questions. | 20 |
| 1 | Which of the following statement is correct for the nitration of benzene?
(a) C-H bond breaking and C-N bond formation occur simultaneously.
(b) Loss of a proton from a Wheland intermediate is very quick and always favoured.
(c) C-H bond breaks in the rate determining step.
(d) C-N bond is formed in a fast step followed by C-H bond breaking in a slow step. | |
| 2 | Which of the following alkanes can exhibit optical activity?
(a) Neopentane (b) 2,3-dimethylbutane
(c) 3-methylhexane (d) 3-methylpentane | |
| 3 | Bromination of cis-2-butene is _____.
(a) single step reaction (b) syn addition (c) antistereoselective (d) all of them | |

- 4 Total number of optically active stereoisomers for 2,3,4-trihydroxypentane is _____.
(a) 2 (b) 4 (c) 6 (d) 8
- 5 In elimination reaction meaning of E1 is _____.
(a) One step elimination (b) elimination of one group
(c) unimolecular elimination (d) none of them
- 6 Which of the following trapping agents can be used to prove the involvement of nitrene as an intermediate?
(a) Benzene (b) Nitroso compounds
(c) Anthracene (d) 2,2,6,6-tetramethylpiperidyl-N-oxyl radical
- 7 Which of the following statement is NOT true for Friedel-Craft acylation reaction?
(a) Polyacylation usually occurs leading to the mixture of products.
(b) It is favoured over Friedel Craft alkylation for synthesizing alkyl benzenes.
(c) Product does not undergo any rearrangement.
(d) More than one mole of Lewis acid catalyst is required for the reaction.
- 8 Which of the following involves carbanion as an intermediate?
(a) Pinacol-Pinacolone rearrangement (b) Demjanov rearrangement
(c) Sommelet-Hauser rearrangement (d) Beckmann rearrangement
- 9 Nitration of nitrobenzene gives _____.
(a) 3-nitroaniline (b) 2,4-dinitrobenzene
(c) 1,4-dinitrobenzene (d) 1,3-dinitrobenzene
- 10 E1cb mechanism involves _____.
(a) simultaneous bond breaking of C-H and C-Y bonds
(b) breaking of C-H bond in a reversible step followed by C-Y bond breaking
(c) irreversible breaking of C-H bond followed by C-Y bond breaking
(d) C-Y bond breaks first then C-H bond breaks
- 11 A bulky base like t-BuOK is used in Darzen Glycidic ester condensation, because _____.
(a) It prevents self condensation.
(b) Weak base cannot abstract proton from substrate.
(c) It prevents the chances for SN2 type displacement in α -bromoester
(d) All of above.

- 12 Nitrobenzene is used as a solvent in Friedel Craft reactions because _____.
 (a) Nitro group deactivates benzene ring and slows down the rate of overall electrophilic substitution.
 (b) It is inert and dissolves AlCl_3 in it.
 (c) It is useful in preventing polyalkylation which often occurs.
 (d) It is responsible for polarization of the reagent.
- 13 Which diester gives 2-carbethoxycyclopentanone as a product by a Dieckmann reaction?
 (a) diethyl malonate (b) diethyl succinate
 (c) diethyl hexanedioate (d) diethyl heptanedioate
- 14 Reaction of benzene with $\text{DCI} / \text{AlCl}_3$ results in to
 (a) Addition of DCI to benzene. (b) Substitution of H by D.
 (c) Substitution of H by Cl. (d) Substitution of D by Cl.
- 15 Involvement of molozonide in ozonolysis of alkene can NOT be confirmed by
 (a) Isolation of molozonide at -70°C . (b) Na/liq.NH_3 reduction to give syn-diol.
 (c) Low temp. NMR spectroscopy. (d) Isolation of molozonide at room temperature.
- 16 In elimination reaction meaning of E_2 is _____.
 (a) Two step elimination (b) elimination of two group
 (c) bimolecular elimination (d) none of them
- 17 Benzene diazonium chloride can undergo diazo coupling with _____.
 (a) anisole (b) benzoic acid (c) salicylic acid (d) m-dinitrobenzene
- 18 Which of the following reaction is a chain extension reaction?
 (a) Curtius Reaction (b) Darzen Glycidic ester condensation
 (c) Beckmann Rearrangement (d) Knoevenagel condensation
- 19 Which of the following method can be used to determine the mechanism of Bayer-Villiger oxidation?
 (a) Primary Kinetic Isotope effect (b) Cross-over experiment
 (c) Isotopic labeling (d) Isotopic exchange
- 20 Following reaction in the presence of Br_2/NaOH is _____.
 $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CO}^{15}\text{NH}_2 + \text{CD}_3\text{-CH}_2\text{-CH}_2\text{-CONH}_2 \longrightarrow \text{CD}_3\text{-CH}_2\text{-CH}_2\text{-}^{15}\text{NH}_2 + \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-NH}_2$
 (a) thermodynamically favoured process (b) reversible process
 (c) intramolecular process (d) intermolecular process

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Date: 29/03/2018 Day: Thursday Time: 10.30 a.m. to 01.00 p.m. Maximum Marks: 50

Important Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
 - Make suitable assumptions and draw neat figures wherever required.
 - Use of non-programmable calculator is allowed.
 - Show necessary calculations.
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SECTION-I

Q-II Answer the following questions as directed.

- A Explain Markovnikov's rule by citing proper example. Discuss the effect of peroxide on reaction. 04
- B Discuss the term "Hyperconjugation" with proper example. 03
- C Define the term Diastereomer and Meso compound with proper example. 03

OR

- C Define the term Enantiomer and Homomer with proper example. 03

Q-III Answer the following questions as directed.

- A Justify: Outcome of the Aldol condensation can be controlled through base strength. 04
- B Explain: Chirogenic centers are always stereogenic but revers is not true. 03
- C Define the term electrophile and nucleophile with proper example. 03

OR

- C Define the term electrofuge and nucleofuge with proper example. 03

SECTION-II

Q-IV Answer the following questions as directed.

- A Give the mechanism for methyl and hydride migration in Wagner-Meerwein rearrangement. 04
- B Discuss the limitation and usefulness of Fischer projection. 03
- C Write the mechanism of Pinacol-pinacolone rearrangement. 03

OR

C Write the mechanism of Cannizzaro reaction. **03**

Q-V Answer the following questions as directed.

A Discuss the bromination of alkene with mechanism. **04**

B Write the mechanism for secondary Beckmann rearrangement. **03**

C Prove the intramolecular migration in Hoffmann rearrangement. **03**

OR

C Prove the intramolecular migration in Curtius rearrangement. **03**

Q-VI Answer the following questions as directed.

A Explain: Friedel-Craft acylation is more advantageous than Friedel-Craft alkylation. **04**

B Write a note on ipso substitution. **03**

C Discuss the electrophilic addition on symmetric diene. **03**

OR

C Discuss the electrophilic addition on asymmetric diene. **03**